Two-Axis Thrust Vector Control Gimbal Demonstrator

Design, Analysis, Fabrication, and Closed-Loop Control

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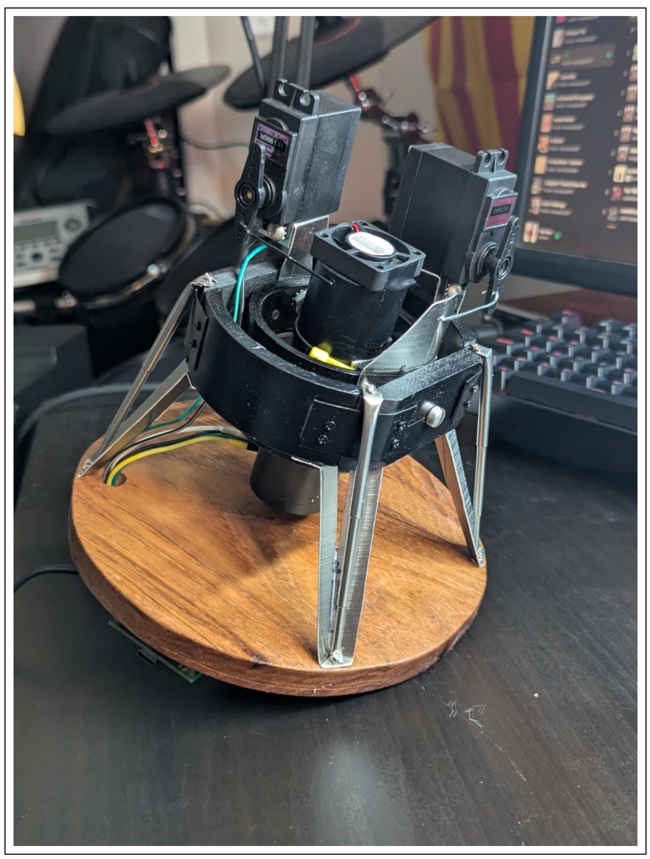
Abstract

This report documents the design, analysis, and construction of a two-axis thrust vector control (TVC) gimbal demonstrator, inspired by the actuated engine mounts used on SpaceX's Raptor engines. The system uses a nested-ring mechanism, driven by two servo linkages, to actively stabilize an inner tube against disturbance. The loop closes with a 6-axis IMU, a complementary filter, and a PID controller running on an ESP32.



Contents

1 Introduction & Conceptual Design **4**
1.1 Motivation . . . 4
1.2 Design Inspiration: Raptor TVC . . . 4
1.3 Initial Sketches . . . 4
1.4 Design Targets . . . 5

2 Constraint & Clearance Analysis **5**
2.1 Geometry Setup . . . 6
2.2 Two-Triangle Derivation, Full-Tip Case . . . 6
2.3 Short-Ring Regime, Full Derivation . . . 7
2.4 Worked Calculation: Inner Ring to Outer Ring . . . 8
2.5 Worked Calculation: Outer Ring to Frame . . . 8
2.6 Confirmed Clearances . . . 9
2.7 Demo Stand Tilt Clearance . . . 9
2.8 Linkage Ratio Linearity Check . . . 9

3 Torque Budget & Actuator Sizing **10**
3.1 Governing Equation . . . 10
3.2 Real Mass Properties . . . 11
3.3 Worked Calculation, Pitch Axis . . . 11
3.4 Worked Calculation, Yaw Axis . . . 12
3.5 Summary . . . 12

4 CAD & Drawing Package **13**
4.1 Parametric Modeling Approach . . . 13
4.2 Sheet Metal Legs and Servo Brackets . . . 14
4.3 GD&T Approach . . . 14
4.4 Drawing Package and Iteration . . . 15

5 Fabrication **16**
5.1 Material Selection: PVC & Hand Fabrication Constraints . . . 16

6 Electronics & Sensor Fusion **17**
6.1 System Architecture . . . 17
6.2 Complementary Filter . . . 17
6.3 Sensor Calibration . . . 17
6.4 Axis Mapping . . . 17

7 Closed-Loop Control **17**
7.1 PID Control: Brief Principles . . . 17
7.2 Anti-Windup . . . 18
7.3 Safety Limit . . . 18
7.4 Final Gains . . . 18
7.5 Pitch/Yaw Divergence: A Late-Stage Debugging Story . . . 19

8 WiFi Live-Tuning Dashboard **20**

9 Circuit Design & PCB **20**
9.1 Switching Circuit Validation . . . 20
9.2 PCB / Carrier Board Approach . . . 21

10 Results **21**
10.1 Step Response . . . 21
10.2 Methodology and Worked Calculation . . . 21
10.3 Per-Event Results . . . 23
10.4 Quantitative Summary . . . 24

11 Iteration & Lessons Learned **25**
11.1 Named Future Improvements . . . 25

12 Bill of Materials **26**

13 Conclusion **27**

A Full Drawing Package **27**

B Firmware Listing **27**

1 Introduction & Conceptual Design

1.1 Motivation

The inspiration for this project ultimately stems from the desire to learn. This, combined with the desire to showcase my engineering skills in a visually compelling portfolio piece, led to the decision to build a two-axis thrust vector control (TVC) gimbal demonstrator. The main inspiration for this project came from the engine mounts and TVC systems used on SpaceX's Raptor engines after seeing the complexities involved and being fascinated by the technology. Not only was this project relevant to recent professional endeavors, but it was interesting to me personally and it was something that I had very little prior knowledge about.

The goal of the project was to showcase as broad of a range of engineering skills as possible. This meant that the project would need to include conception, mechanical design, electrical design, control systems, and software development. The combination of these skills allows for the demonstration of a complete system, from concept to implementation. Along with this, the project was also designed to be visually compelling and hopefully inspire others to continue learning and exploring the world of engineering.

1.2 Design Inspiration: Raptor TVC

The mechanism design draws directly from the actuated engine mounts used on SpaceX's Raptor engines (albeit a different mechanism), which gimbal the entire engine assembly to steer thrust vector direction rather than relying on separate control surfaces.

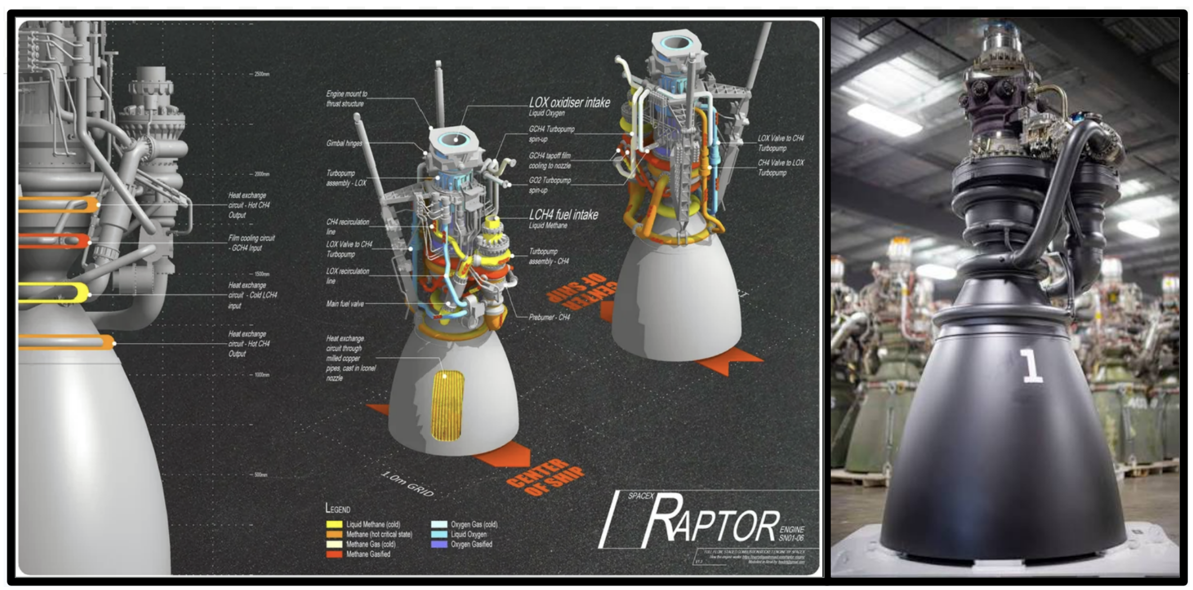


Figure 1: SpaceX Raptor engine [1], the inspiration for this project's actuated gimbal mechanism.

1.3 Initial Sketches

Initial conception began with hand sketches, exploring the nested-ring mechanism, linkage geometry, and overall layout. These sketches are largely conceptual and clearly do not resemble the final project geometry. However, they were essential in the brainstorming phase and helped to establish the overall design direction.

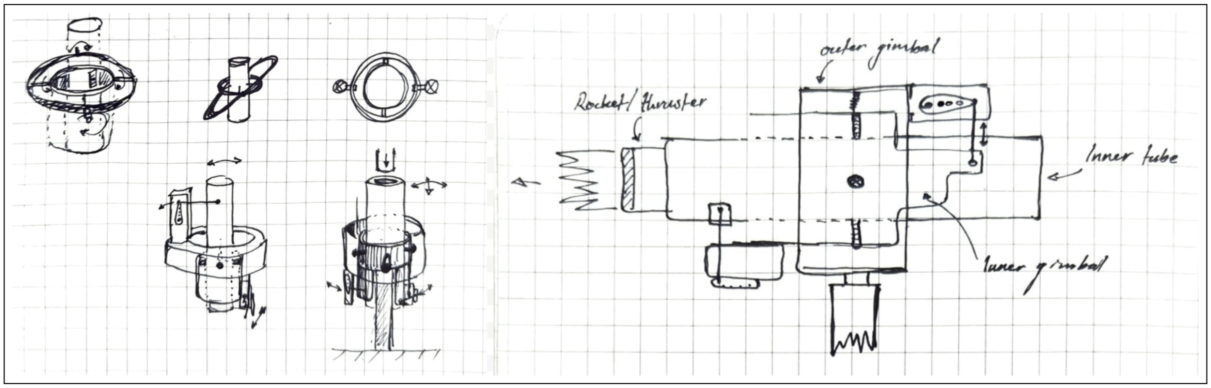


Figure 2: Initial concept page, July 7, 2026.

1.4 Design Targets

The following constraints were fixed at the outset of the project and used throughout all subsequent clearance and torque calculations:

- Per-axis deflection: $\theta = 15°$
- Combined worst-case deflection (both axes simultaneously): $\theta_c = \sqrt{2} \times 15° = 21.21°$
- Fabrication margin: 3 on all clearance checks
- Linkage ratio: 2:1 (servo rotation : gimbal deflection)

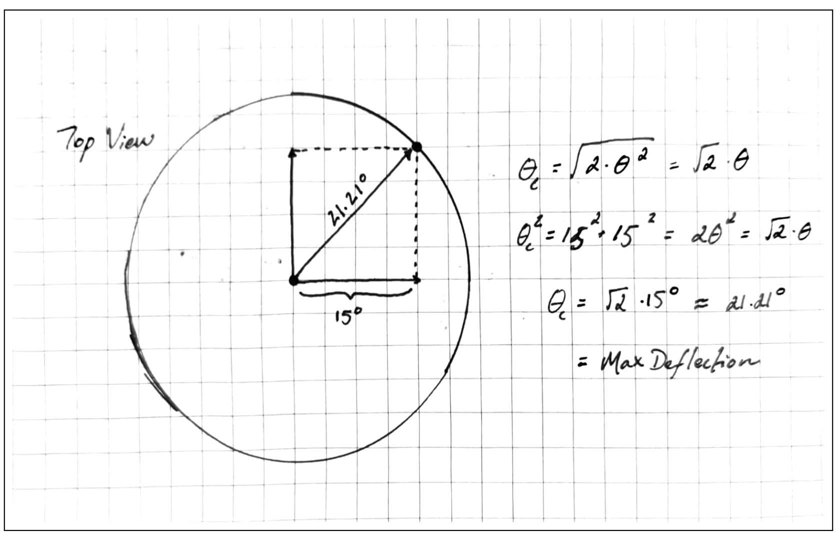


Figure 3: Combined-deflection derivation for $\theta = 15°$.

2 Constraint & Clearance Analysis

This section derives, from first principles, the geometric clearance required between each pair of nested rings as they rotate through their full range of motion.

2.1 Geometry Setup

Each ring pair is modeled as a rigid arm pivoting about a fixed point, decomposed into two perpendicular legs: an axial offset h and a radial offset r. Rotating this arm by angle θ requires resolving both legs' contributions to the resulting radial reach independently.

2.2 Two-Triangle Derivation, Full-Tip Case

Consider a ring pivoting about a fixed point, with radius r_2 (its outer edge) and half-height h (more specifically,its axial extent from the pivot to its end). At rest, the straight-line distance from the pivot to the ring's farthest corner is the hypotenuse of a right triangle with legs r_2 and h:

$$L_e = \sqrt{r_2^2 + h^2} \tag{1}$$

This hypotenuse sits at a fixed angle φ from the axial (vertical) direction, determined entirely by the ring's own proportions (this is the furthest radial reach the ring's corner can achieve, hence the h being 1/2 the ring's total height):

$$\varphi = \arctan\left(\frac{r_2}{h}\right), \qquad \sin\varphi = \frac{r_2}{L_e}, \qquad \cos\varphi = \frac{h}{L_e} \tag{2}$$

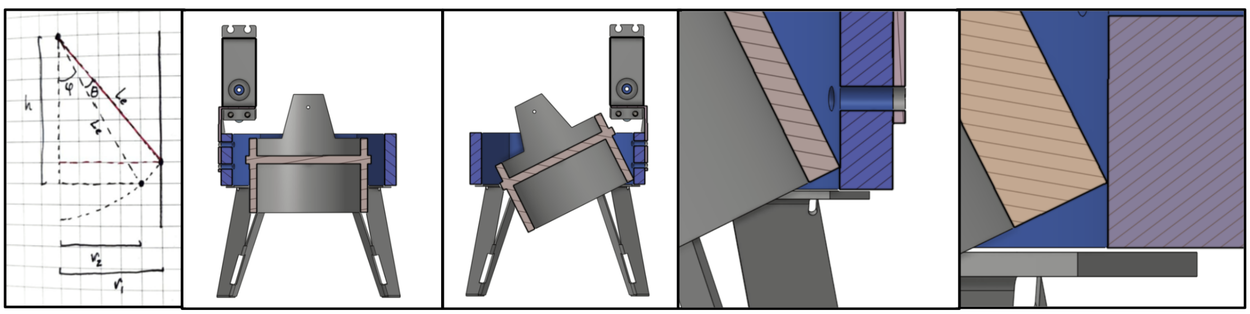


Figure 4: Full-tip geometry: pivot, radius r_2, half-height h, hypotenuse L_e at rest angle φ, rotated by θ.

When the ring rotates by the design deflection angle θ, the corner's hypotenuse now sits at angle $\varphi + \theta$ from the axial direction (red traced hypotenuse). The radial distance this corner reaches from the pivot axis is:

$$r_1 = L_e \sin(\varphi + \theta) \tag{3}$$

Expanding with the angle-sum identity:

$$\sin(\varphi + \theta) = \sin\varphi\cos\theta + \cos\varphi\sin\theta \tag{4}$$

Substituting the expressions for $\sin\varphi$ and $\cos\varphi$ found above:

$$r_1 = L_e \left[\frac{r_2}{L_e}\cos\theta + \frac{h}{L_e}\sin\theta\right] \tag{5}$$

The L_e terms cancel cleanly, leaving a result that depends only on the ring's own radius and half-height, not on the hypotenuse itself:

$$r_1 \geq r_2 \cos\theta + h \sin\theta + \text{margin} \tag{6}$$

This is the full-tip clearance requirement: it gives the minimum inner radius a containing ring must have for a rotating ring's farthest corner to clear it at deflection θ.

2.3 Short-Ring Regime, Full Derivation

If the containing ring is short enough that the rotating ring's corner sweeps past its edge before reaching full radial extension, the full-tip formula above no longer applies. Instead, the constraint becomes: the containing ring's own corner, swung *backward* by θ, must just clear the rotating ring's surface at radius r_2.

Set up the same way, but now using the containing ring's own dimensions r_{out} and h_{out}:

$$L_w = \sqrt{r_{\text{out}}^2 + h_{\text{out}}^2}, \qquad \varphi_w = \arctan\left(\frac{r_{\text{out}}}{h_{\text{out}}}\right), \qquad \sin\varphi_w = \frac{r_{\text{out}}}{L_w}, \qquad \cos\varphi_w = \frac{h_{\text{out}}}{L_w} \tag{7}$$

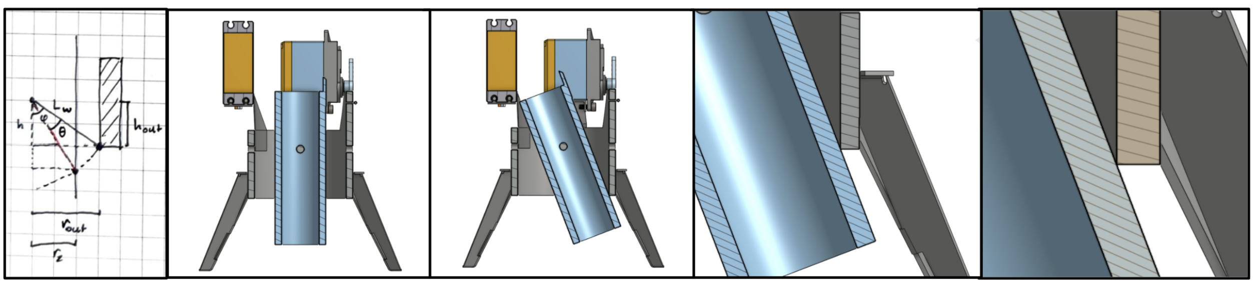


Figure 5: Short-ring geometry: hypotenuse L_w from the containing ring's own corner, rest angle φ_w, rotated backward by θ.

The governing condition is that this corner, rotated backward by θ, lands exactly at r_2:

$$L_w \sin(\varphi_w - \theta) = r_2 \tag{8}$$

Expanding:

$$\sin(\varphi_w - \theta) = \sin\varphi_w \cos\theta - \cos\varphi_w \sin\theta \tag{9}$$

Substituting:

$$r_2 = L_w \left[\frac{r_{\text{out}}}{L_w}\cos\theta - \frac{h_{\text{out}}}{L_w}\sin\theta\right] = r_{\text{out}}\cos\theta - h_{\text{out}}\sin\theta \tag{10}$$

Solving for r_{out}:

$$r_{\text{out}}\cos\theta = r_2 + h_{\text{out}}\sin\theta \tag{11}$$

$$r_{\text{out}} \geq \frac{r_2}{\cos\theta} + h_{\text{out}}\tan\theta + \text{margin} \tag{12}$$

This gives the minimum radius a *short* containing ring needs, correctly accounting for the fact that the rotating ring's corner passes the containing ring's edge before reaching its full theoretical radial extent.

The calculator automatically selects between Equation 6 and Equation 12 based on each ring pair's actual proportions.

2.4 Worked Calculation: Inner Ring to Outer Ring

Inputs: inner ring OD = 33.47 mm (1" nominal, measured), so $r_2 = 16.735$ mm. Inner ring half-height $h = 50$ mm. Deflection $\theta = 15°$: this interface uses the single-axis pitch deflection, not the combined worst case, since yaw rotates the outer ring and everything inside it (including the inner tube) together as a single unit and creates no relative tilt between inner tube and outer ring. Margin = 3 mm.

Step 1, naive full-tip estimate. Applying Equation 6 directly, before checking which regime actually governs:

$$r_{1,\text{naive}} = r_2 \cos\theta + h \sin\theta + \text{margin} = 16.735(0.9659) + 50(0.2588) + 3 = 32.11 \text{ mm} \tag{13}$$

Step 2, check the actual regime. Compute L_e and φ:

$$L_e = \sqrt{r_2^2 + h^2} = \sqrt{16.735^2 + 50^2} = 52.73 \text{ mm}, \qquad \varphi = \arctan\left(\frac{r_2}{h}\right) = 18.51° \tag{14}$$

The real outer ring stock's half-height, $h_{\text{outer}} = 30$ mm, is shorter than the inner ring's own half-height (50 mm). This means the inner ring's tip, when swung by θ, clears past the outer ring's own end before reaching full radial extension, so the *short-ring* regime (Equation 12) governs instead of the naive full-tip case.

Step 3, correctly-regime-selected requirement.

$$r_{\text{outer}} \geq \frac{r_2}{\cos\theta} + h_{\text{outer}} \tan\theta + \text{margin} = \frac{16.735}{0.9659} + 30(0.2679) + 3 = 28.36 \text{ mm} \tag{15}$$

Step 4, check against real stock. The selected outer ring (2-1/2" nominal coupling) has a measured ID of 69.06 mm, giving $r_{1,\text{actual}} = 30.325$ mm:

$$\text{Clearance} = 30.325 - 28.36 = 1.96 \text{ mm} \quad \checkmark \text{ clears} \tag{16}$$

2.5 Worked Calculation: Outer Ring to Frame

Inputs: outer ring's own radius $r_{2,\text{outer}} = 34.53$ mm, half-height $h_{\text{outer}} = 30$ mm. Frame half-height $h_{\text{frame}} = 15$ mm. Deflection $\theta = 15°$: this interface uses the single-axis yaw deflection, since pitch is internal to the outer ring and does not tilt the outer ring relative to the frame at all.

Step 1, check regime. The frame ($h_{\text{frame}} = 15$ mm) is now shorter than the outer ring's own half-height ($h_{\text{outer}} = 30$ mm) - the opposite situation from the full-tip case. The short-ring regime governs here.

Step 2, radial requirement.

$$r_{\text{frame}} \geq \frac{r_{2,\text{outer}}}{\cos\theta} + h_{\text{frame}} \tan\theta + \text{margin} = \frac{34.53}{0.9659} + 15(0.2679) + 3 = 42.77 \text{ mm} \tag{17}$$

Frame's actual radius (from a measured 44.63 mm internal socket dimension) gives:

$$\text{Clearance} = 44.63 - 42.77 = 1.86 \text{ mm} \quad \checkmark \text{ clears} \tag{18}$$

Axial check: not applicable. With $h_{\text{frame}} = 15$ mm now shorter than $h_{\text{outer}} = 30$ mm, the frame no longer axially encloses the outer ring's full extent even at rest - it functions as a radial support band around the tube's middle rather than a full enclosure. The axial-enclosure check used when the frame was the taller member no longer applies to this geometry and is omitted.

2.6 Confirmed Clearances

Table 1: Ring-pair clearance results, each interface at its own governing single-axis deflection ($\theta = 15°$)

Interface	Required (mm)	Actual (mm)	Margin (mm)	Governing regime
Inner ring → Outer ring (radial)	28.36	30.325	1.96	Short-ring
Outer ring → Frame (radial)	42.77	44.63	1.86	Short-ring

2.7 Demo Stand Tilt Clearance

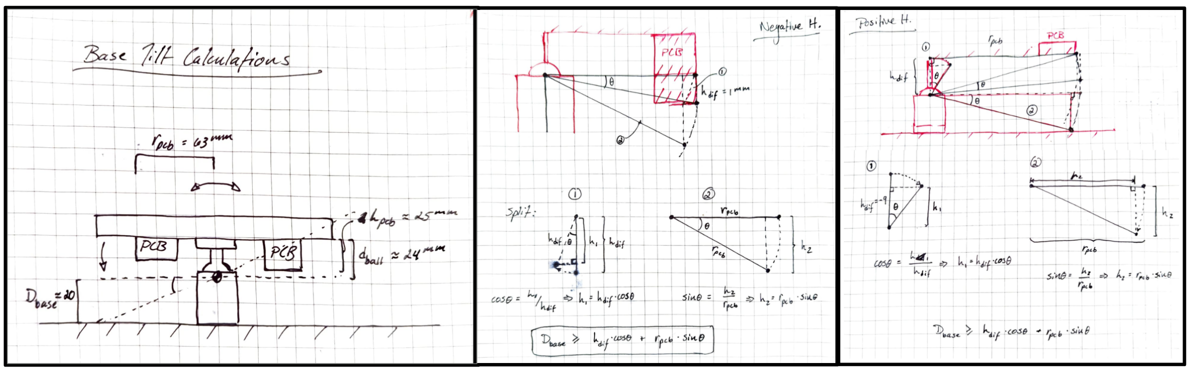


Figure 6: Demo stand tilt clearance geometry, showing the electronics assembly's lowest point h_{pcb} relative to the ball-head pivot d_{ball}, and the required vertical clearance D_{base} to the stationary base plate. Right two images shoing demo stand tilt calculations where the difference in height between the base plate and pcb is negative, and positive, respectively.

The same two-triangle method was reused to size the demo stand's legs, this time solving for the vertical clearance D_{base} required between the ball-head pivot and the stationary base plate as the whole assembly tilts by θ_{demo}:

$$D_{\text{base}} \geq v \cos\theta_{\text{demo}} + r_{\text{pcb}} \sin\theta_{\text{demo}} + \text{margin} \tag{19}$$

where $v = d_{\text{ball}} - h_{\text{pcb}}$ is the signed vertical offset of the electronics assembly's lowest point relative to the pivot. Two populated PCBs were checked independently; the governing (larger) required clearance determined final leg length. Along with mathematical calculations, a simulated check was done by modeling this and in Onshape and visually confirming by using mate features that the assembly could tilt to the required angle without interference.

2.8 Linkage Ratio Linearity Check

The 2:1 servo-to-gimbal linkage ratio assumes the servo arm's small angular sweep translates linearly into gimbal deflection. This assumption was checked directly by comparing arc length against chord length at the design deflection of $\theta = 15°$:

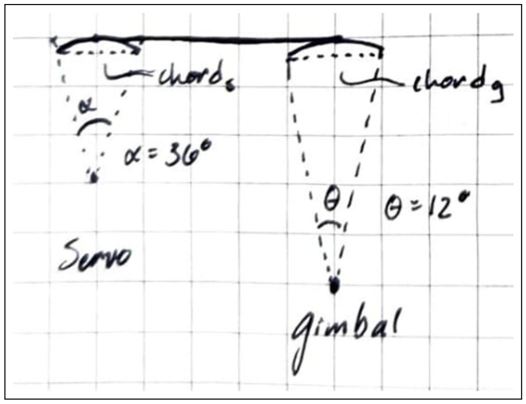


Figure 7: Sketch of linkage ratio geometry, where theta is 12 instead of 15 in this case (previous constraint).

Gimbal side (independent of linkage ratio):

$$\text{Arc: } S_a = 0.26180r \tag{20}$$
$$\text{Chord: } S_c = 2r\sin(7.5^\circ) = 0.26105r \tag{21}$$
$$\text{Error} = \frac{S_a - S_c}{S_a} \approx 0.29\% \tag{22}$$

Servo side, at the actual 2:1 ratio (servo angle $= 2 \times 15^\circ = 30^\circ$):

$$\text{Arc: } S_a = 0.52360r \tag{23}$$
$$\text{Chord: } S_c = 2r\sin(15^\circ) = 0.51764r \tag{24}$$
$$\text{Error} = \frac{S_a - S_c}{S_a} \approx 1.14\% \tag{25}$$

Both errors are small enough that the linear small-angle approximation was adopted for the linkage geometry without further correction, particularly given the system operates closed-loop, where any small residual geometric nonlinearity is continuously corrected by the PID controller rather than accumulating open-loop.

3 Torque Budget & Actuator Sizing

3.1 Governing Equation

Each axis's required torque was budgeted from three contributions:

$$\tau_{\text{gravity}} = mg\, d_{\text{cg}} \sin\theta \tag{26}$$
$$\tau_{\text{thrust}} = FL\sin\theta \tag{27}$$
$$\tau_{\text{inertial}} = I\alpha, \quad I = m\, d_{\text{cg}}^2, \quad \alpha = \frac{4\theta_{\text{rad}}}{t^2} \tag{28}$$

with a 1.30 friction margin applied and divided by the 2:1 linkage ratio to reach the required servo torque:

$$\tau_{\text{servo}} = \frac{1.30(\tau_{\text{gravity}} + \tau_{\text{thrust}} + \tau_{\text{inertial}})}{2} \tag{29}$$

3.2 Real Mass Properties

Mass properties were pulled directly from Onshape (not estimated), using mate-connector-referenced centers of mass for each axis's actual moving assembly. Distance from pivot to center of mass, d_{cg}, was computed directly from the three-axis offset reported by Onshape:

$$d_{\text{cg}} = \sqrt{X^2 + Y^2 + Z^2} \tag{30}$$

Pitch (inner tube assembly): $X = 0.364\,\text{mm}$, $Y = 0.44\,\text{mm}$, $Z = -7.144\,\text{mm}$, mass $= 66.582\,\text{g}$

$$d_{\text{cg,pitch}} = \sqrt{0.364^2 + 0.44^2 + 7.144^2} = 7.167 \text{ mm} = 0.007167 \text{ m} \tag{31}$$

Yaw (outer tube assembly): $X = -2.948\,\text{mm}$, $Y = -14.721\,\text{mm}$, $Z = 14.363\,\text{mm}$, mass $= 206.125\,\text{g}$

$$d_{\text{cg,yaw}} = \sqrt{2.948^2 + 14.721^2 + 14.363^2} = 20.777 \text{ mm} = 0.020777 \text{ m} \tag{32}$$

Other inputs, held constant across both axes: thrust reaction force $F = 0.15\,\text{N}$ (fan), moment arm to fan $L = 0.038\,\text{m}$ (measured directly from the real pivot-to-fan geometry, 101 mm total minus 63 mm pivot-to-tube-base), target settle time $t = 0.2\,\text{s}$, friction margin 30%, deflection $\theta = 15° = 0.2618\,\text{rad}$.

3.3 Worked Calculation, Pitch Axis

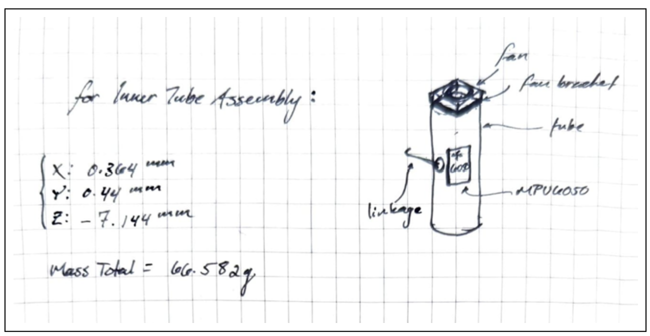


Figure 8: Pitch axis torque budget sketch, showing center of gravity.

$$\begin{aligned}
\tau_{\text{gravity}} &= m\,g\,d_{\text{cg}}\sin\theta &&= 0.066582 \times 9.81 \times 0.007167 \times \sin(15°) = 1.212 \times 10^{-3} && (33)\\
\tau_{\text{thrust}} &= FL\sin\theta &&= 0.15 \times 0.038 \times \sin(15°) = 1.475 \times 10^{-3} && (34)\\
I &= m\,d_{\text{cg}}^2 &&= 0.066582 \times (0.007167)^2 = 3.421 \times 10^{-6} && (35)\\
\alpha &= \frac{4\theta}{t^2} &&= \frac{4 \times 0.2618}{(0.2)^2} = 26.18 && (36)\\
\tau_{\text{inertial}} &= I\alpha &&= 3.421 \times 10^{-6} \times 26.18 = 8.96 \times 10^{-5} && (37)
\end{aligned}$$

$$\tau_{\text{subtotal}} = 0.001212 + 0.001475 + 0.0000896 = 0.002777 \text{ N·m} \tag{38}$$

$$\tau_{\text{total}} = 1.30 \times 0.002777 = 0.003610 \text{ N·m} \tag{39}$$

$$\tau_{\text{servo}} = \frac{0.003610}{2} = 0.001805\ \text{N·m} \tag{40}$$

$$\text{Safety factor} = \frac{0.92}{0.001805} \approx 510\times \tag{41}$$

3.4 Worked Calculation, Yaw Axis

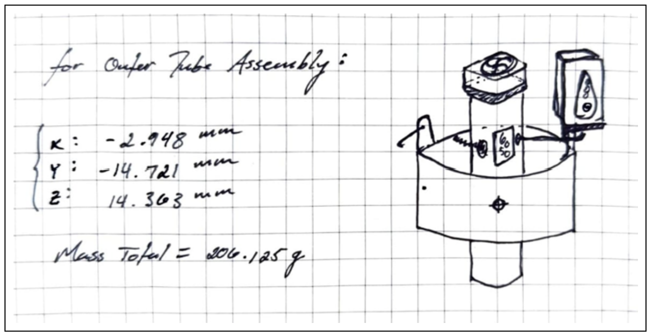


Figure 9: Yaw axis torque budget sketch, showing center of gravity.

$$\begin{aligned}
\tau_{\text{gravity}} &= m\,g\,d_{\text{cg}}\sin\theta = 0.206125 \times 9.81 \times 0.020777 \times \sin(15^\circ) = 1.0875 \times 10^{-2} && (42)\\
\tau_{\text{thrust}} &= FL\sin\theta \qquad = 0.15 \times 0.038 \times \sin(15^\circ) = 1.475 \times 10^{-3} \quad \text{(same } F,\ L,\ \theta \text{ as pitch)} && (43)\\
I &= m\,d_{\text{cg}}^2 \qquad = 0.206125 \times (0.020777)^2 = 8.899 \times 10^{-5} && (44)\\
\alpha &= \frac{4\theta}{t^2} \qquad = 26.18 \quad \text{(same } \theta,\ t \text{ as pitch)} && (45)\\
\tau_{\text{inertial}} &= I\alpha \qquad = 8.899 \times 10^{-5} \times 26.18 = 2.330 \times 10^{-3} && (46)
\end{aligned}$$

$$\tau_{\text{subtotal}} = 0.010875 + 0.001475 + 0.002330 = 0.014680\ \text{N·m} \tag{47}$$

$$\tau_{\text{total}} = 1.30 \times 0.014680 = 0.019084\ \text{N·m} \tag{48}$$

$$\tau_{\text{servo}} = \frac{0.019084}{2} = 0.009542\ \text{N·m} \tag{49}$$

$$\text{Safety factor} = \frac{0.92}{0.009542} \approx 96\times \tag{50}$$

3.5 Summary

Both axes carry substantial margin against the selected MG996R servo's rated torque (0.92 at 4.8), an intentional choice given the servo's low incremental cost relative to the risk of an undersized actuator.

Table 2: Torque budget results, both axes

Axis	Mass (kg)	d_{cg} (m)	Safety Factor
Pitch	0.066582	0.007167	$\approx 510\times$
Yaw	0.206125	0.020777	$\approx 96\times$

4 CAD & Drawing Package

4.1 Parametric Modeling Approach

For this project, the online software Onshape was chosen to parametrically model the design. This would allow for easy modification and iteration of the design. Specifically, key dimensions (leg length, mounting offsets) were maintained as named Onshape variables rather than hard-coded values, allowing late-stage geometry changes (e.g. leg length, once real PCB height was known) to propagate automatically rather than requiring manual rework. Using this software also facilitated the createion of a full drawing package, including a comprehensive bill of materials (BOM) and revision table, which was useful in tracking changes throughout the project. The full assembly is visible in Figure 10 and is publicly available at the following url: https://cad.onshape.com/documents/38454ed41460ebb3f5376222/w/7e6b36a949ced50e5d7d3c83/e/bf2871f1439507b37564f52c

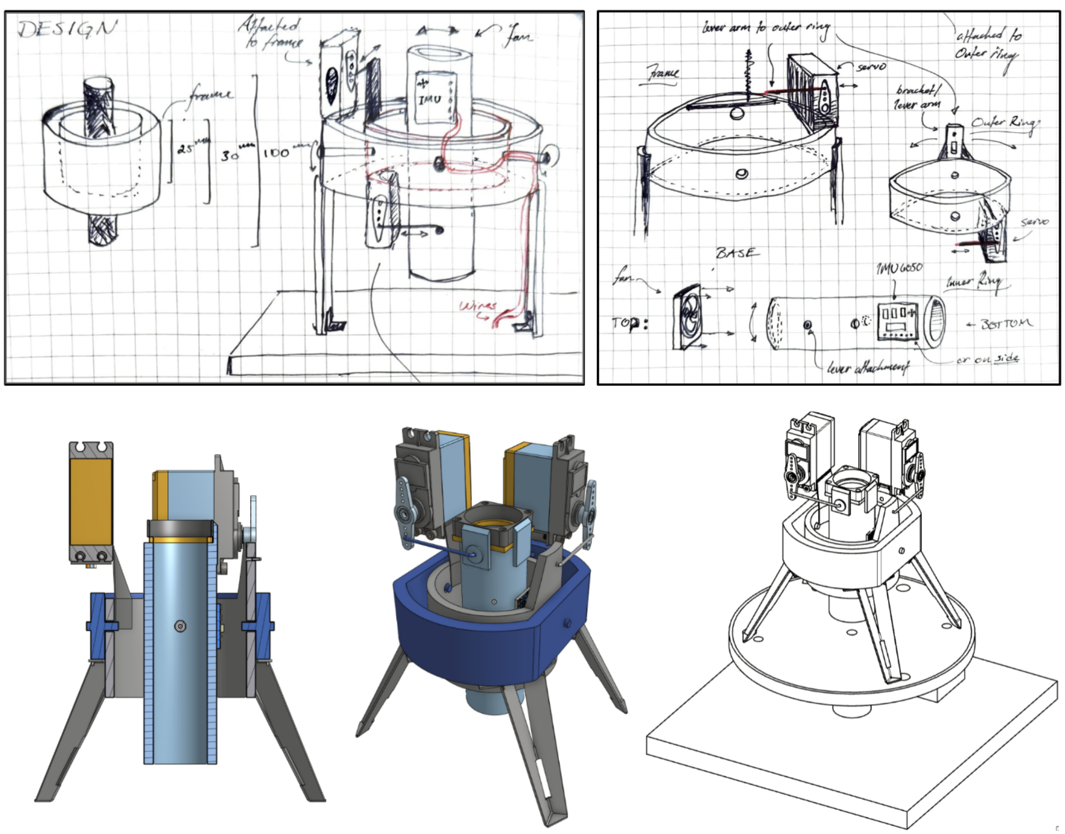


Figure 10: Full assembly design sketch, showing frame, outer ring, base, and wiring routing with corresponding cad model and drawing.

4.2 Sheet Metal Legs and Servo Brackets

The demonstration stand's legs combine a purchased telescoping antenna (upper section) with a hand-formed aluminum V-channel (lower section), modeled in Onshape's Sheet Metal workflow to generate an accurate flat pattern for hand fabrication 11(A-D). The servo brackets were also modeled in Onshape but, because of the complex double axis bend, the software was not able to generate a true sheet metal pattern. Therefore, an alternate method was used instead: a pen and ruler 11(E-F). All sheet metal parts were fabricated from 0.8 mm brushed aluminum sheet stock and were formed using a combination of hand tools and a small bench vise.

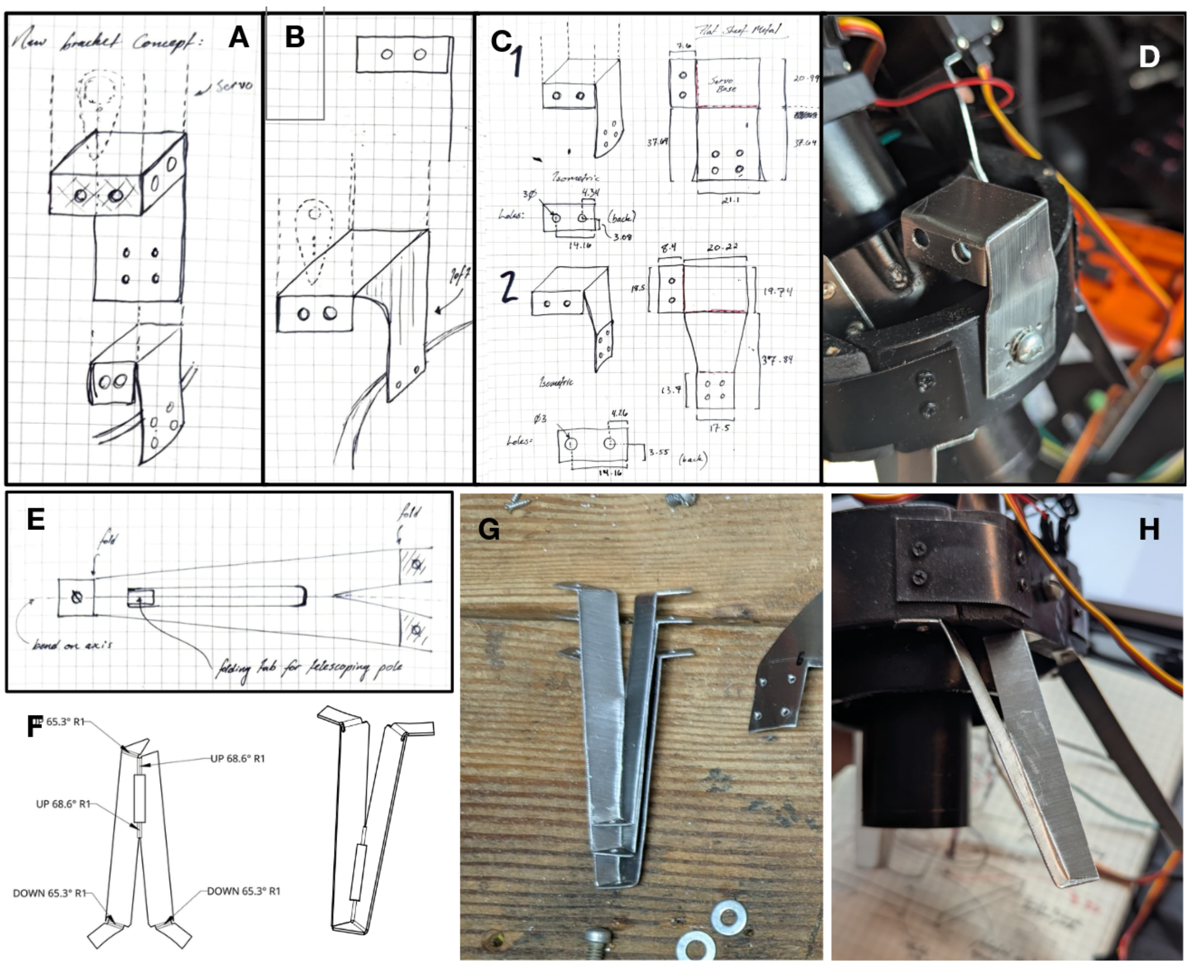


Figure 11: A-D) Sheet metal servo bracket design sketches, templates, and hand-fabricated implementation. E-F) Sheet metal leg design sketches, templates, and hand-fabricated implementation.

4.3 GD&T Approach

Given hand fabrication (not machined production), full GD&T was deliberately not applied. Instead, the following tolerances were noted:

- Bend angles: ±2° tolerance, called out explicitly given bending is the least repeatable hand operation in the build.
- Linear dimensions: ±0.5 mm general tolerance.

4.4 Drawing Package and Iteration

Figure 10 depicts the process of creating the full assembly design sketch, showing the initial conceptualizations of features such as the frame geometry, servo placement, and imu placement. However, the final design was iterated upon and refined from the conceptual phase. Namely, the frame was flattented on either side so as to avoide the requirement for spacers and to allow for a more compact and structurally sound design. The servo placement was also iterated upon and it can be seen in the figure that both servos were moved to the same side of the assembly as opposed to oposing sides as seen in the conceptual sketches. This was done to allow for a more compact design overall, as the bottom mounted servo would have required a larger gap between the frame and the base plate due to the extension past the lowest part of the inner tube.

The sensor placement was also changed from the conceptual sketches, as the IMU unit was moved to the center of the inner tube. This was done in hopes to eliminate any potential gravitational bias that may be introduced by the IMU being mounted outside of the pivot point. In theory, in this configuration, the IMU bias on the yaw axis should be negligible and the pitch axis bias is minimized for a mounting configuration that isn't inside of the inner tube. Most of the iteration progress described above can be seen in Figure 12.

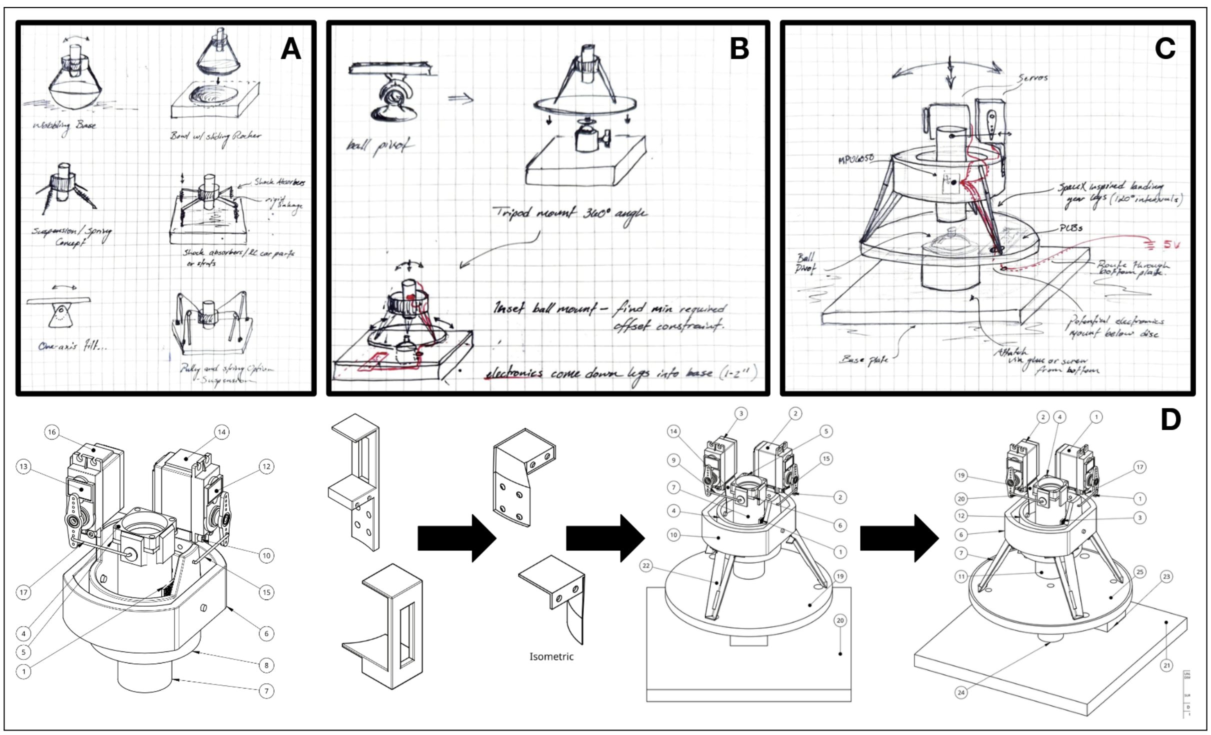


Figure 12: A) Conceptual base plate design sketches; B) Ball point pivot for base plate concept; C) Full base plate concept V.01 incorporating a base plate and connected ball point pivot joint for full range of motion; D) Assembly revision progression, showing evolution of the brackets, legs, and base plate geometry.

5 Fabrication

5.1 Material Selection: PVC & Hand Fabrication Constraints

A self imposed budget of \$100 was set in place at the beginning of the project, which limited material selection to low cost, commercially available materials. After brainstorming, PVC was selected as the primary material for the inner and outer tube assemblies, as well as the frame. PVC was low cost, easy to machine by hand, durable, and readily available in a variety of sizes.

Given no access to a 3D printer or machine shop, all custom parts were designed around this commercially available catalog stock (PVC pipe/couplings, aluminum sheet) and hand tools. This constraint is treated throughout this report as a documented design decision rather than a limitation; see Section 11 for named improvements that would be pursued given access to additional fabrication tools.

The inner tube was selected to be 1" nominal PVC pipe, while the outer tube was selected to be 2-1/2" nominal PVC coupling. The frame was selected to be 3" nominal PVC coupling (Figure 13). The final design was iterated upon to ensure that the inner tube assembly would clear the outer tube assembly, and that the outer tube assembly would clear the frame, at the maximum deflection angle of 15 degrees. The final design also ensured that the inner tube assembly would clear the base plate at the maximum deflection angle of 15 degrees. These calculations can be seen in Section 2

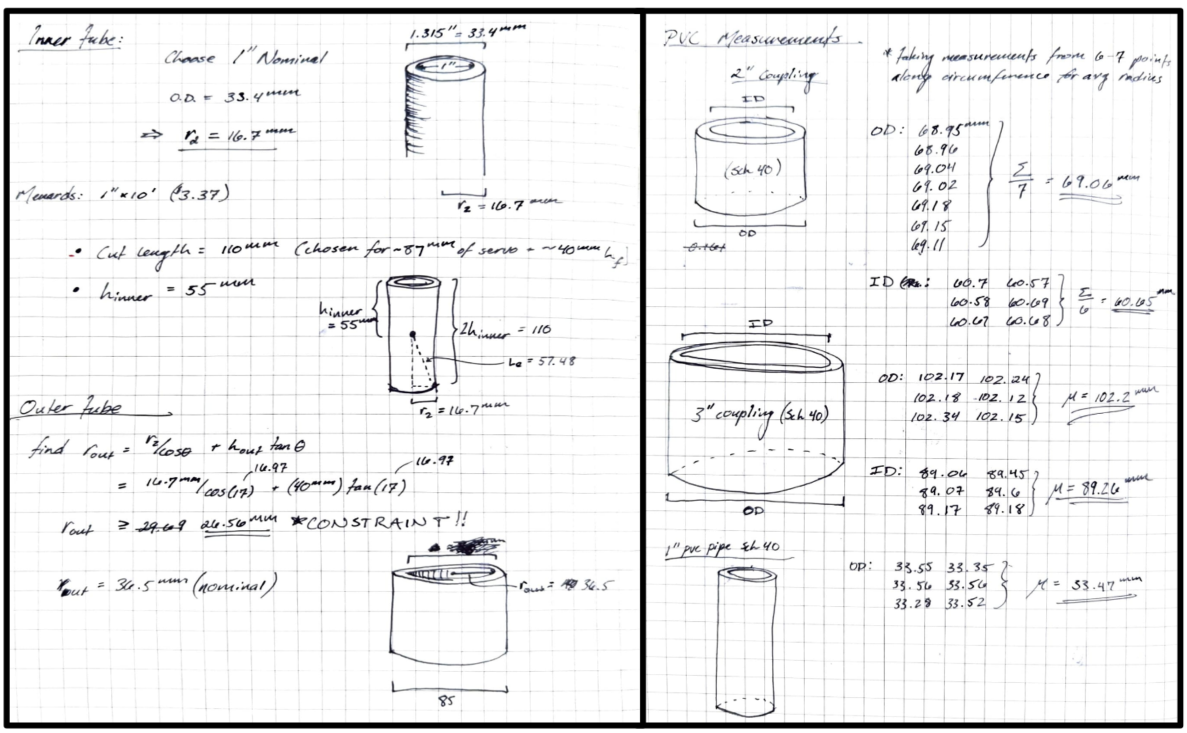


Figure 13: Inner/outer tube sizing derivation using real catalog dimensions and PVC caliper measurements, 1" pipe and 2"/3" couplings,averaged over 6-7 points around each circumference.

6 Electronics & Sensor Fusion

6.1 System Architecture

An ESP32 (WROOM-32D) reads a 6-axis IMU (MPU6050) over I2C, computes a fused orientation estimate, and drives two servos through a PCA9685 PWM driver, also communicating over I2C and sharing the same bus at a different address.

6.2 Complementary Filter

Orientation is estimated by fusing gyroscope integration (fast, drift-prone) with an accelerometer-derived tilt angle (slow, but drift-free), following the standard complementary filter approach [2]:

$$\theta_{\text{fused}} = \alpha(\theta_{\text{fused}} + \dot{\theta}_{\text{gyro}}\,\Delta t) + (1-\alpha)\,\theta_{\text{accel}} \tag{51}$$

with α typically in the 0.95–0.998 range depending on axis.

6.3 Sensor Calibration

Six offsets (three accelerometer, three gyroscope) were determined by averaging 300 samples with the sensor held stationary in its final mounted orientation, and re-derived after every physical remount. Final offsets used in the firmware are as follows:

Table 3: Final IMU offset values (calibration averages, final mounted orientation)

Axis	Accelerometer offset (g)	Gyroscope offset (deg/s)
X	−0.9616	0.5768
Y	−0.0555	1.3094
Z	−0.0584	−0.6862

6.4 Axis Mapping

As previously stated, the IMU's final mounting position is on the side of the inner tube, with the axis orientations as seen in Figure 14. As will be discussed later in the report, there were a number of issues with the two axes' behavior when tuning the PID values. Full troubleshooting details can be found in the README on the github repo for the project. For now, this gives us a concrete mapping that we can now input the gyroscope and accelerometer data into which will feed into the PID loop and dictate our servo behavior.

7 Closed-Loop Control

7.1 PID Control: Brief Principles

A standard PID control law was used per axis:

$$u(t) = K_p\,e(t) + K_i\int_0^t e(\tau)\,d\tau + K_d\frac{de(t)}{dt} \tag{52}$$

where $e(t)$ is the fused orientation error relative to a level setpoint. The proportional term provides an immediate, error-proportional correction; the integral term eliminates steady-state

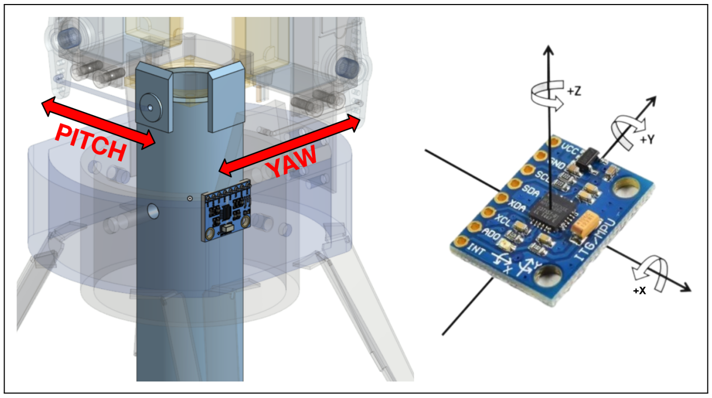


Figure 14: IMU axis mapping sketch, showing the IMU's physical orientation on the inner tube and the corresponding raw axis mapping to pitch and yaw.

error under sustained disturbance (a genuinely necessary term for this application, since the stand is tilted and *held* during demonstration, not just briefly disturbed); the derivative term damps oscillatory overshoot.

7.2 Anti-Windup

A conditional-integration anti-windup scheme was implemented: the integral term only accumulates when the commanded output is not already saturated at the mechanical safety limit, or when the current error would pull the output back into range. This was a necessary correction after an earlier, simpler fixed-ceiling integral clamp allowed the controller to remain saturated for extended periods during large disturbances, producing a slow, overshooting recovery once the disturbance was removed.

7.3 Safety Limit

All commanded corrections are hard-clamped to a maximum gimbal deflection of 15°, independent of the calculated PID output. This value is set to match the per-axis deflection target established in Section 2 and used throughout every clearance and torque calculation in this report, rather than being an independently chosen firmware constant. This guarantees the firmware can never command the mechanism beyond the deflection range the physical design was actually validated for. However, while this did prevent any catostrophic movements, this value can be raised to 20° for a more visually striking demonstration.

7.4 Final Gains

Table 4: Final PID gains

Axis	K_p	K_i	K_d
Pitch	-0.6	-6.0	-0.0001
Yaw	-0.6	-15.0	-0.001

Final complementary filter blend: pitch 0.65 gyro / 0.35 accelerometer; yaw 0.9 gyro / 0.1 accelerometer.

7.5 Pitch/Yaw Divergence: A Late-Stage Debugging Story

Despite running structurally identical control code on both axes, pitch persistently exhibited behavior yaw never did: jitter at rest, edge oscillation at a naive hard deadband boundary, an integral windup jolt, and a brief wrong-direction servo flick at the onset of any fast tilt. Each of the first three had clean, confirmed root causes: a discontinuous hard deadband (fixed by making the deadband continuous, subtracting the threshold from the error rather than gating output on and off), integral accumulation continuing silently during the deadband freeze (fixed by the same continuous-deadband change, since the error itself now reaches zero rather than being externally gated), and a stale gyroscope offset macro left pointing at an outdated calibration constant after a later recalibration.

The remaining flick proved more stubborn. A first-principles hypothesis was pursued in depth: the IMU is mounted 0.1674 m from the pitch pivot but essentially at the yaw pivot, so real centripetal ($l\omega^2$) and tangential ($l\alpha$) acceleration should appear at the sensor during pitch rotation and not during yaw. This effect is well documented in the IMU/gimbal literature and was confirmed as a genuinely real phenomenon. In practice, the centripetal and tangential acceleration terms were computed directly from angular velocity and angular acceleration, then subtracted from the raw accelerometer reading before the tilt angle was calculated:

$$a_c = l\,\omega^2, \qquad a_t = l\,\alpha, \qquad \alpha = \frac{d\omega}{dt} \tag{53}$$

where l is the measured distance from the pivot to the IMU. Both terms were converted to units of g and subtracted from the relevant raw accelerometer axis before computing the corrected tilt angle:

$$a_{x,\mathrm{corr}} = a_x - \frac{l\,\omega^2}{g}, \qquad a_{z,\mathrm{corr}} = a_z - \frac{l\,\alpha}{g} \tag{54}$$

```
float omega = gyroRate * PI / 180.0;
float alpha = (omega - lastOmega) / dt;

float centripetal_g = (L * omega * omega) / G;
float tangential_g  = (L * alpha) / G;

float axCorrected = ax - centripetal_g;
float azCorrected = az - tangential_g;

float accelPitchCorrected = atan2(azCorrected,
    sqrt(axCorrected * axCorrected + ay * ay)) * 180.0 / PI;
```

Despite the rigor, implementing and tuning it (correct axis assignment, rate smoothing to control derivative noise, several attempts at gating the correction's influence) never fully resolved the symptom, only reshaped it.

The eventual fix was structurally simpler than the physics model being pursued: reverting to a stripped-down control loop with zero correction terms, identical structure on both axes, and directly retuning the complementary filter's blend ratio. Counter to the working assumption throughout the physics investigation, pitch required *more* accelerometer trust than yaw, not less, once the blend ratio was allowed to move freely rather than being computed indirectly through a motion-based correction term. This is treated in this report as a genuine finding rather than a loose end: the sophisticated, individually well-justified physical correction was solving a real effect that was not, in practice, the dominant contributor to the observed symptom, and a much simpler direct retuning outperformed it.

8 WiFi Live-Tuning Dashboard

An asynchronous web server, hosted directly on the ESP32, exposes a lightweight dashboard for adjusting all six PID gains live, without reflashing, along with a polling telemetry readout of current angle and servo position. This proved essential for iterative tuning, since safety limits (the hard 15° clamp) remain enforced in firmware regardless of what values are entered. However, as with everything nice in this world, it did come with some tradeoffs. The live `\trace` endpoint builds a JSON string from several arrays via repeated string concatenation. This operation is apparently very taxing on the ESP32 architecture and polling this every ∼150ms while the dashboard is open steals enough loop time to visibly deteriorate the performance. This results in jumpy servo behavior that does not dissipate unless the dashboard is closed. In order to complete the results section under the system's highest performance, the dashboard was set to start recording for a preset amount of time (∼20 seconds) before it could be closed and reopened after the test was complete.

9 Circuit Design & PCB

9.1 Switching Circuit Validation

The fan's PWM speed-control circuit (NPN transistor low-side switch) was validated in NI Multisim prior to physical construction, confirming correct base resistor sizing, transistor saturation behavior, and PWM waveform integrity via simulated oscilloscope measurement. Full simulation process notes are maintained separately in the project README.

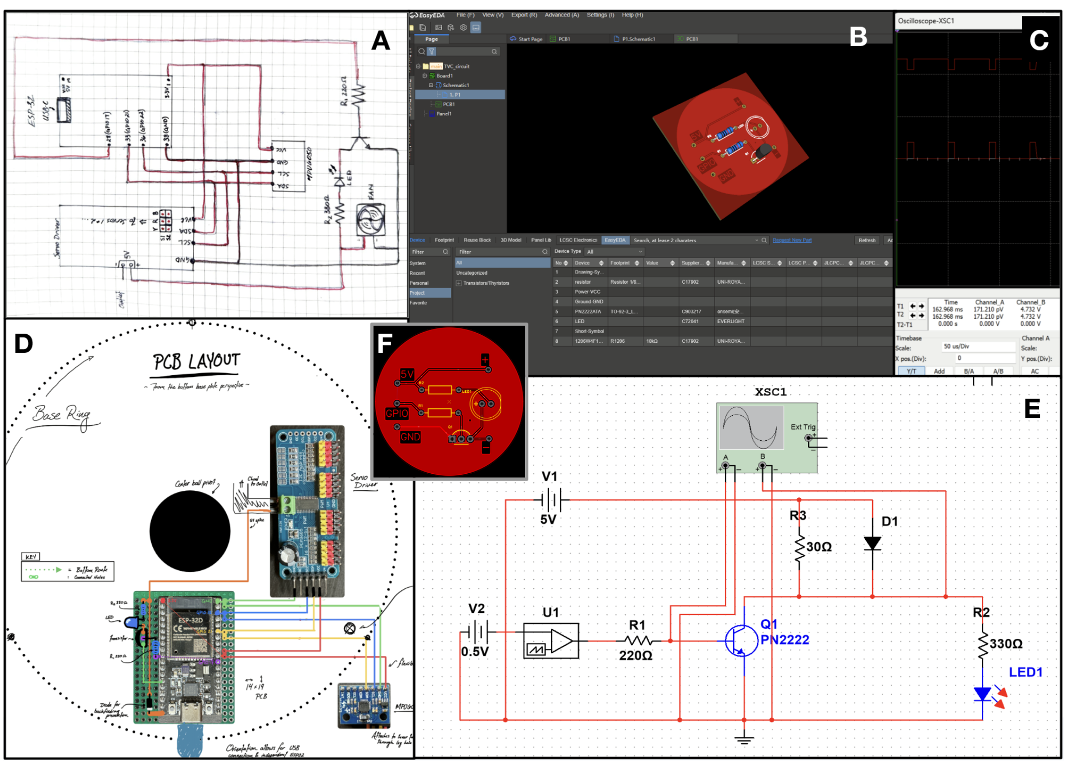


Figure 15: FILL

9.2 PCB / Carrier Board Approach

Given that the majority of the system's electronics are pre-built breakout modules (ESP32, PCA9685, IMU) rather than raw components, the custom PCB was designed as a connector/header carrier board, providing mounting and wiring consolidation discrete fan-switching components, rather than a fully custom circuit. However, due to the limited project budget, the PCB was not physically implemented and the designed circuit was implemented into the breakout board, as seen in Figure 15.

10 Results

Step-response data was captured directly from the running system via a recording buffer on the ESP32 (timestamped at the actual loop rate, ∼24 ms per sample), triggered manually for each test session and downloaded as JSON for offline analysis. Two dedicated single-axis sessions are used here: one focused entirely on yaw disturbances, one entirely on pitch, each spanning approximately 24 seconds and containing multiple distinct, manually-applied disturbances. Settling time is defined as the time from disturbance onset until the angle enters and remains within a $\pm 2°$ band of its pre-disturbance baseline; steady-state error is the final angle relative to that baseline once settled.

10.1 Step Response

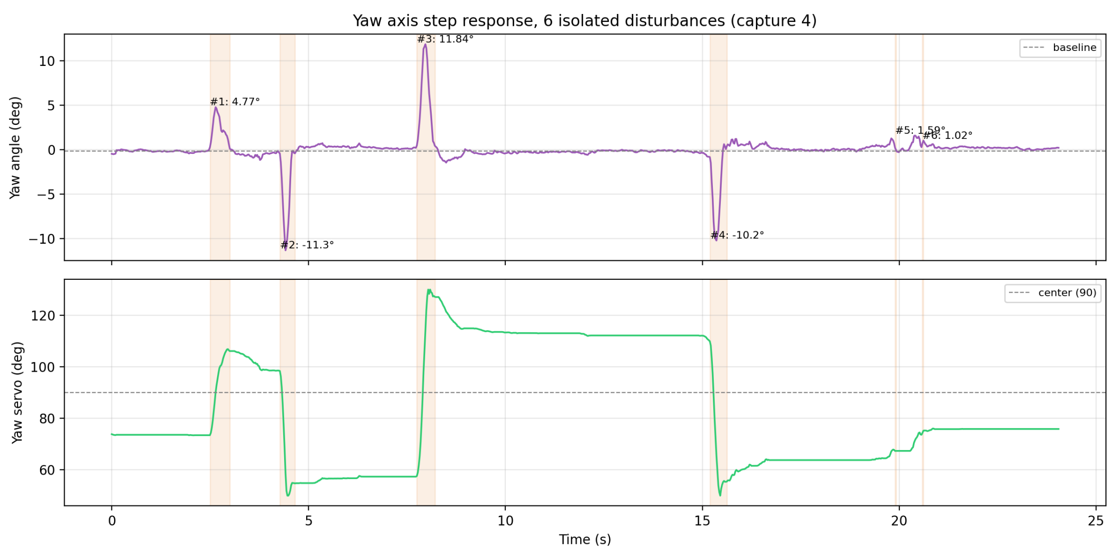


Figure 16: Yaw axis angle and servo output across 6 isolated disturbances. All six settled cleanly within the tolerance band.

10.2 Methodology and Worked Calculation

Four quantities are computed per disturbance event: peak deflection, true overshoot, settling time, and steady-state error. All depend on first establishing a baseline - the resting angle immediately before a given disturbance begins, computed as the mean of the angle signal over the quiet interval preceding it:

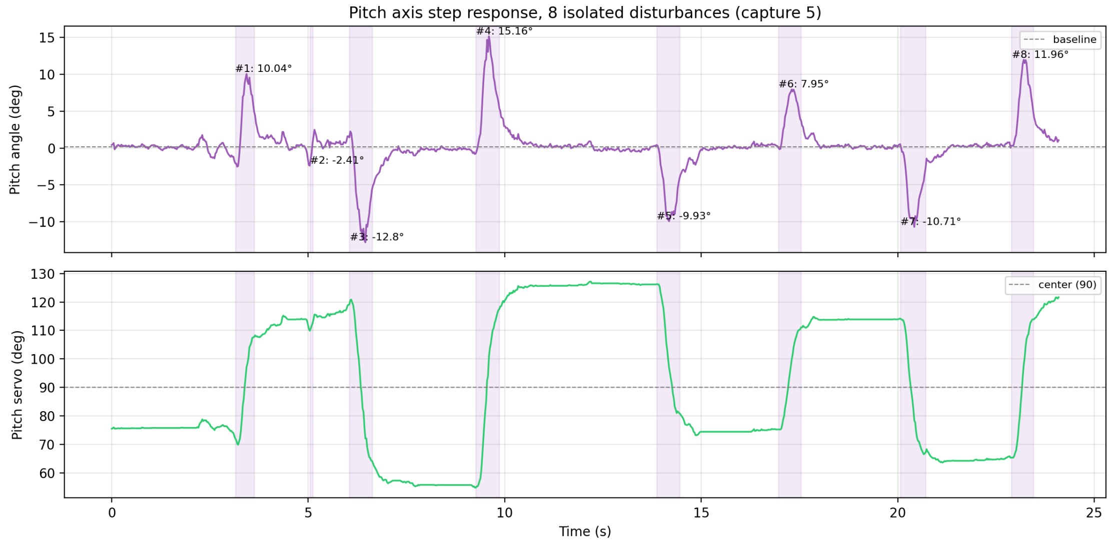


Figure 17: Pitch axis angle and servo output across 8 isolated disturbances. Seven of eight settled cleanly; event 1 did not settle before the next disturbance began.

$$\beta = \frac{1}{n}\sum_{i=1}^{n}\theta_i, \qquad i \in [\text{quiet interval}] \tag{55}$$

Worked example, yaw baseline. Using the 89 samples recorded before the first yaw disturbance:

$$\beta_{\text{yaw}} = \frac{1}{89}\sum_{i=1}^{89}\theta_i = -0.179^\circ \tag{56}$$

For each event, the search window runs from the disturbance's onset to the start of the *next* disturbance (or the end of the recording, for the final event). Peak deflection is the largest deviation from baseline anywhere in that window - this describes the size of the manually-applied disturbance itself, not the controller's response to it:

$$\theta_{\text{peak}} = \theta_i \text{ such that } |\theta_i - \beta| \text{ is maximized}, \qquad i \in [\text{search window}] \tag{57}$$

Overshoot is distinct from peak deflection. Peak deflection measures how far the disturbance pushed the platform, which is a property of the applied disturbance, not the controller. True overshoot measures the controller's own behavior during correction: after the peak, does the angle cross back past the baseline and swing to the *opposite* side before settling, and if so by how much. This is computed by scanning the signal from the peak onward and taking the largest deviation on the opposite side of baseline from the disturbance itself:

$$\text{Overshoot} = \max_{i > i_{\text{peak}}} \left[-\operatorname{sign}(\theta_{\text{peak}} - \beta) \cdot (\theta_i - \beta) \right] \tag{58}$$

taken with the sign of the crossing direction. An event that decays monotonically back to baseline without ever crossing to the opposite side has zero true overshoot.

Settling time is the first moment, measured from disturbance onset, after which the angle remains within a $\pm 2^\circ$ tolerance band around β for the remainder of the search window:

$$t_{\text{settle}} = t_j - t_{\text{onset}}, \quad j = \min\{i : |\theta_k - \beta| \le 2^\circ \ \forall k \ge i \text{ within window}\} \tag{59}$$

Steady-state error is the angle at the end of the search window, relative to baseline:

$$e_{ss} = \theta_{\text{window end}} - \beta \tag{60}$$

Worked example, yaw event 3. Disturbance onset at $t = 7.749\,\text{s}$; next disturbance begins at $t = 15.189\,\text{s}$. Peak deflection occurs at $t = 7.965\,\text{s}$, $\theta_{\text{peak}} = 11.837^\circ$ (disturbance size $= 11.837 - (-0.181) = 12.02^\circ$, i.e. the size of the applied push - not the reported overshoot). Scanning forward from the peak, the angle crosses baseline and reaches a minimum of $\theta = -1.45^\circ$ before settling:

$$\text{Overshoot} = -1.45 - (-0.18) = -1.27^\circ \tag{61}$$

The angle first enters and remains within the $\pm 2^\circ$ band at $t = 8.157\,\text{s}$:

$$t_{\text{settle}} = 8.157 - 7.749 = 0.408 \text{ s} \tag{62}$$

At the end of the search window, $\theta = -0.836^\circ$:

$$e_{ss} = -0.836 - (-0.181) = -0.656^\circ \tag{63}$$

Worked example, pitch event 4. Pitch's largest recorded disturbance. Onset at $t = 9.263\,\text{s}$, baseline $\beta_{\text{pitch}} = 0.151^\circ$, next disturbance at $t = 13.871\,\text{s}$. Peak deflection at $t = 9.599\,\text{s}$, $\theta_{\text{peak}} = 15.162^\circ$ (disturbance size 15.01°). After the peak, the angle crosses baseline and reaches $\theta = -0.53^\circ$ before settling:

$$\text{Overshoot} = -0.53 - 0.151 = -0.68^\circ \tag{64}$$

Settles at $t = 10.151\,\text{s}$:

$$t_{\text{settle}} = 10.151 - 9.263 = 0.888 \text{ s} \tag{65}$$

At window end, $\theta = 0.418^\circ$:

$$e_{ss} = 0.418 - 0.151 = 0.267^\circ \tag{66}$$

This same procedure was applied identically to all fourteen recorded events (six yaw, eight pitch); full per-event results are given below.

10.3 Per-Event Results

Table 5: Yaw axis, per-event results (6 events, all settled)

Event	Peak (deg)	Overshoot (deg)	Settling time (s)	Steady-state error (deg)
1	4.77	−0.98	0.41	−0.25
2	−11.30	0.91	0.29	0.66
3	11.84	−1.27	0.41	−0.66
4	−10.20	1.45	0.31	0.39
5	1.59	0.00	0.00	0.90
6	1.02	0.00	0.00	0.39

Table 6: Pitch axis, per-event results (8 events, 7 settled)

Event	Peak (deg)	Overshoot (deg)	Settling time (s)	Steady-state error (deg)
1	10.04	N/A (did not settle)	did not settle	−2.56
2	−2.41	2.32	0.14	1.29
3	−12.80	0.11	0.89	−0.90
4	15.16	−0.68	0.89	0.27
5	−9.93	0.73	1.03	−0.11
6	7.95	−0.33	0.67	0.37
7	−10.71	0.63	0.82	0.15
8	11.96	0.00	0.86	0.91

10.4 Quantitative Summary

Table 7: Measured performance vs. design targets (settled events only)

Metric	Target	Yaw (measured)	Pitch (measured)
Max deflection	15	11.84	15.16
Settling time (mean)	–	0.24	0.76
Settling time (range)	–	0.00–0.41	0.14–1.03
Overshoot (mean)	–	0.77	0.69
Overshoot (max)	–	1.45	2.32
Steady-state error (mean)	–	0.54	0.57

With overshoot correctly measured as post-correction crossing past baseline rather than the size of the applied disturbance, both axes show tight, comparable control: yaw's mean overshoot (0.77) and pitch's (0.69) are essentially equivalent, both well under 1° from disturbances as large as 10-15°. Pitch's single worst case (2.32, event 2) exceeds yaw's worst case (1.45, event 4), but this is a single outlier rather than a systematic difference between the axes. Steady-state error is likewise comparable (0.54 vs. 0.57) - both settle to within roughly 1.5× the 0.35 deadband width, close to the practical precision floor of this control scheme.

The one metric that does show a clear, repeatable gap is settling time: yaw settled cleanly on all six recorded disturbances at a mean of 0.24, while pitch settled cleanly on seven of eight at a mean roughly 3× longer (0.76), and its one unsettled event (event 1) never resolved before the next disturbance arrived. This is consistent with the final gain values themselves - yaw runs at roughly double pitch's integral gain ($K_i = -16$ vs. $K_i = -7$) - and with the divergence investigation in Section 7.5: pitch's final complementary filter blend (0.65 gyro / 0.35 accelerometer) leans on the accelerometer roughly 3.5× more heavily than yaw's (0.9/0.1). The accelerometer was established earlier in this report as the noisier, more motion-corrupted of the two signals, so pitch's angle estimate is structurally built from a less clean input than yaw's - plausibly the source of its slower settling, even though this same noise does not translate into meaningfully worse overshoot or final precision once the response is correctly measured.

11 Iteration & Lessons Learned

A detailed, chronological debugging log (sensor calibration, PID tuning, firmware dependency issues) is maintained separately in the project README and referenced here rather than reproduced in full.

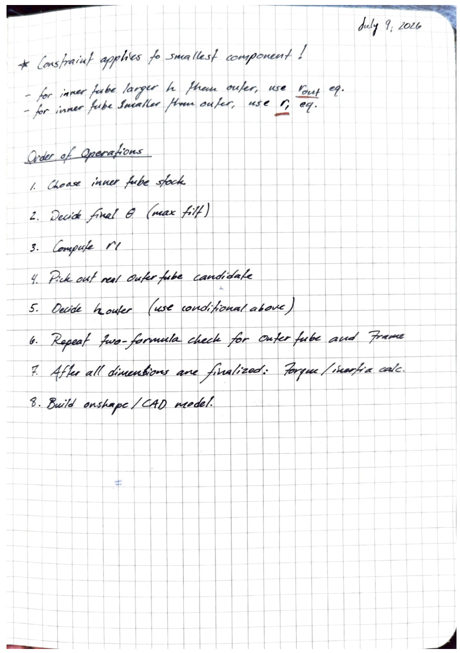
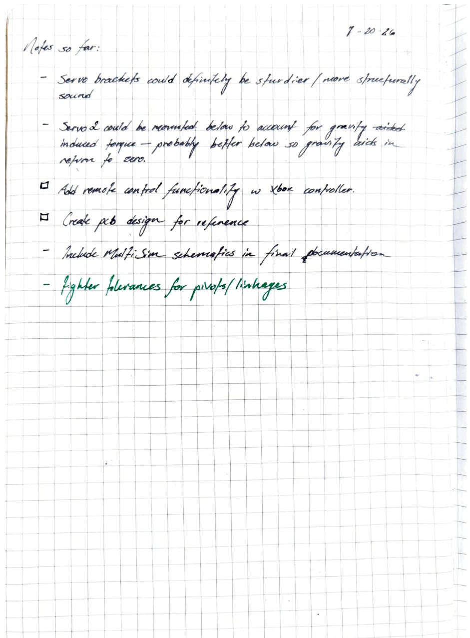


Figure 18: Order of operations, July 9, 2026, establishing the sequence used throughout the geometric design process.

11.1 Named Future Improvements

- **Servo brackets**: current hand-cut brackets are functional but would benefit from 3D-printed replacements for improved rigidity, contingent on tool access.
- **Sensor mounting**: a more deliberate initial axis-alignment process (rather than iterative remount-and-recalibrate) would reduce setup time on any future build.
- **IMU position**: relocating the IMU to sit at (or very near) the true geometric center, equidistant from both the pitch and yaw pivots, would give both axes yaw's current structural advantage (near-zero lever arm) rather than just one. This would likely eliminate the motion-induced centripetal/tangential acceleration error at the mechanical level, removing the need for any software correction rather than continuing to compensate for it after the fact.
- **Yaw axis backward-motion overshoot**: yaw exhibits slight jitter/overshoot specifically when the servo moves in the backward direction, not the forward direction. The leading hypothesis is asymmetric mass distribution (the servo's own weight sitting off to one side of the yaw axis), giving the assembly different effective inertia depending on direction of travel. Not yet confirmed via direct testing; a mechanical counterweight or direction-dependent damping tuning are both candidate fixes.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Notes so far:

- Servo brackets could definitely be sturdier / more structurally sound
- Servo 2 could be mounted below to account for gravity ~~aided~~ induced torque — probably better below so gravity aids in return to zero.
- ☐ Add remote control functionality w Xbox controller.
- ☐ Create pcb design for reference
- Include MultiSim schematics in final documentation
- tighter tolerances for pivots/linkages

Figure 19: Notes so far, July 20, 2026: a dated punch list of known improvements and remaining tasks.

12 Bill of Materials

Name	Material	Description	Mass	Qty	Part Number	Vendor
Servo 1	N/A	MG996R 55g digital metal gear servo	55 g	1	B07MFK266B	Amazon
Servo 2	N/A	MG996R 55g digital metal gear servo	55 g	1	B07MFK266B	Amazon
MPU6050 module	PCB - Fiberglass	6-axis accelerometer/gyroscope, I2C, immersion gold PCB (generic, ICGOICIC)	1.8 g	1	ICGOICIC	AliExpress
Fan	ABS	3007 (30×30×7mm) 5V DC cooling fan	1.881 g	1	701715553996	Amazon
Fan Bracket	Aluminum 6061	Custom fab	2.553 g	1	–	–
Frame	PVC	Custom fab	82.711 g	1	–	–
Gimbal Leg	Aluminum	Custom fab	4.161 g	4	–	–
Inner Tube	PVC	Custom fab	59.597 g	1	–	–
Outer Ring	PVC	Custom fab	62.097 g	1	–	–
Screw, DIN 912 M3×8 A2	Hardened carbon steel	Socket head cap screw, A2 stainless	0.876 g	4	91292A112	McMaster-Carr
Servo1 Bracket	Aluminum 6061	Custom fab	4.58 g	1	–	–
Servo1 Linkage	Stainless steel	Custom fab	0.454 g	1	–	–
Servo2 Bracket	Aluminum 6061	Custom fab	2.621 g	1	–	–
Servo2 Linkage	Stainless steel	Custom fab	0.645 g	1	–	–
Base Plate	Basswood	Custom fab	304.92 g	1	–	–

Name	Material	Description	Mass	Qty	Part Number	Vendor
Base Plate Disc	Basswood	Custom fab	105.887 g	1	–	–
Base Plate Disc*	Basswood	Custom fab	100.192 g	1	–	–
ESP32	PCB - Fiberglass	WROOM-32D dev board, dual-core, WiFi+Bluetooth, USB-C	9 g	1	ESP32-DevKitC-32	Amazon
Ball Joint Pivot	N/A	Mini tripod ball head, 360° pivot	175.371 g	1	–	–
Servo Driver	PCB - Fiberglass	PCA9685 16-channel 12-bit PWM driver, I2C	26.9 g	1	B07RMTN4NZ	Amazon

*Two Base Plate Disc entries with differing mass are present in the source CAD model; this likely reflects either two genuinely distinct disc parts or a stale duplicate from an earlier revision, and should be resolved directly in Onshape before this table is treated as final.

13 Conclusion

A Full Drawing Package

B Firmware Listing

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